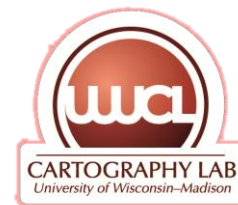


Unit 5: Design Sprint 3 | 13.05.25 14:30–17:00+
Polishing Your Visual Story

Schedule at a Glance



Day 1 // Monday 12.05.25

- **Unit 1: Introductions & Inspiration Talks**
 - 10:00 to 10:15 // Arrival (and coffee!)
 - 10:15 to 10:30 // Introductions from Georg Gartner and Robert Roth
 - 10:30 to 10:45 // Inspiration Talk 1: The United Nations Sustainable Development Goals (Ayako Kagawa)
 - 10:45 to 11:00 // Inspiration Talk 2: Tips for Mapping the SDGs (Robert Roth)
 - 11:00 to 11:15 // Inspiration Talk 3: Making an Atlas in a Day (Alicia Cowart)
- 11:15 to 11:30 // Break
- **Unit 2: Team Formation**
 - 11:30 to 12:00 // Picking Your SDG (rank your top three SDG themes, narrow into small groups, facilitated by Robert Roth)
 - 12:00 to 13:00 // Brainstorming Your Story (breakouts with Alicia Cowart, Merve Keskin, N.T. Nawshin, Robert Roth, Zdenek Stachon, and available TUW staff)
- 13:00 to 14:00 // Lunch
- **Unit 3: Design Sprint 1, Obtaining Your Data, Researching Your Topic**
 - 14:00 to 15:30 // Team Work (organizers available for questions)
- 15:30 to 16:00 // Creative Break, the Marshmallow Challenge (facilitated by Robert Roth)
- **Unit 4: Design Sprint 2, Sketching & Organizing Your Visual Story**
 - 16:00 to 17:30 // Team Work (organizers available for questions) *The goal is to have cleaned, mappable data in a GIS by day's end*
 - 17:30+ // Continue Work As Needed (Optional)

Day 2 // Tuesday 13.05.25

- **Unit 5: Design Sprint 3, Polishing Your Visual Story**
 - 14:30 to 15:00 // Adobe Illustrator Demo
 - 15:00 to 17:00 // Team Work (organizers available for questions) *The goal is to be out of GIS and into Adobe Illustrator by day's end*
 - 17:00+ // Continue Work As Needed (Optional)

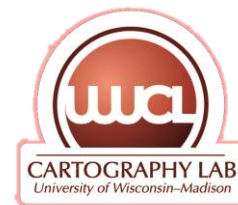
Day 3 // Wednesday 14.05.25

- Initial Drafts Due in Google Slides by 16:00
- **Unit 6: Team Presentations & Draft Feedback**
 - 16:00 to 18:00 // Presentations
- 18:00+ // Social (!) *Recommendations appreciated!*

Day 4 // Thursday 15.05.25

- Second Drafts Due to Rob by 16:00
- PhD Student Show and Tell (N.T. Nawshin, Gareth Baldrice-Franklin) *Contact Rob if you are interested in giving a 5-10 minute lightning talk*
- 18:00+ // PhD Student Social (Alicia and Rob will crash!) *Recommendations appreciated!*

Schedule at a Glance



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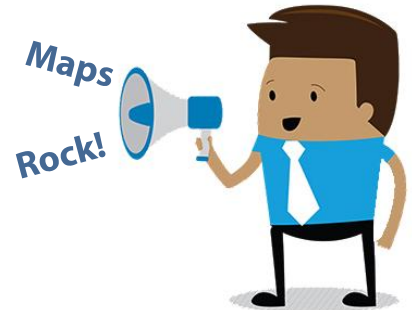
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Unit 5 Objectives

1. Demonstrate tips and tricks for polishing mapping in graphic design software
2. Critique cartographic design decisions that are difficult to change outside of GIS
3. Explore different visual styles that reinforce an SDG story according to the form, color, type, and texture
4. Describe the creative design process behind an SDG atlas spread as an artistic statement
5. Develop an SDG atlas spread in graphic design software using the grid

Important: The goal is to be out of GIS and into Adobe Illustrator by day's end!

Important: Take a Day 2 progress screenshot; final draft screenshot should be added to Google Slides by **16:00 tomorrow** (Wednesday)

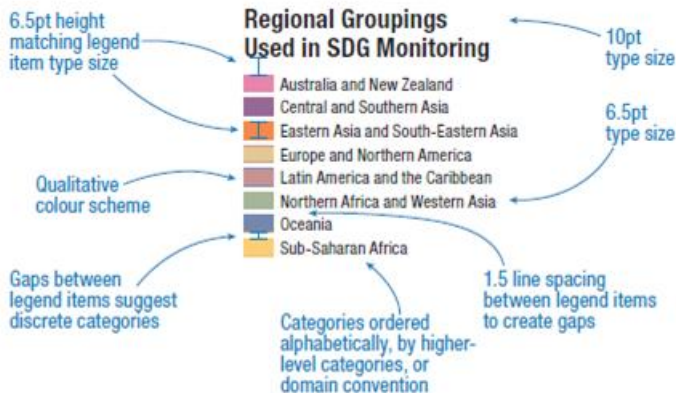
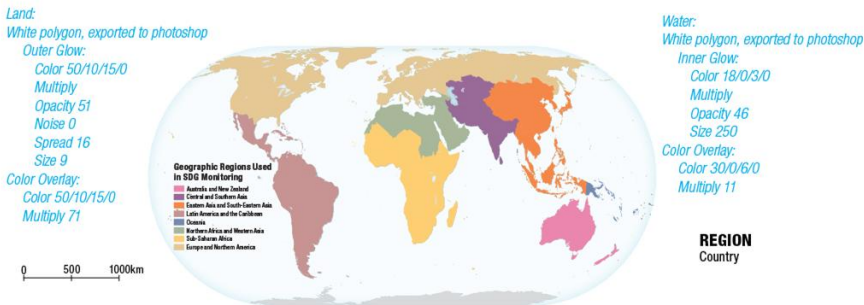


STROKES & CORNERS

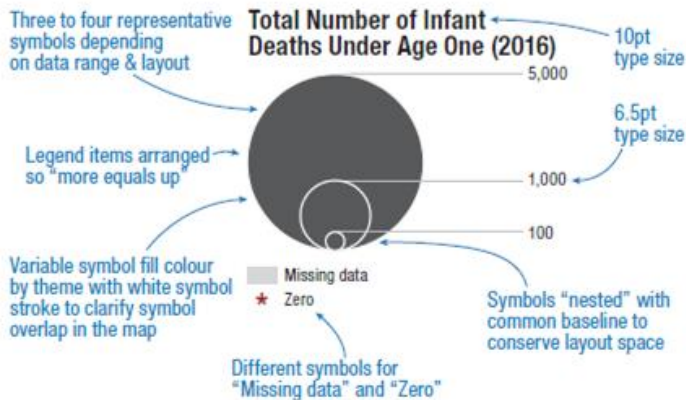
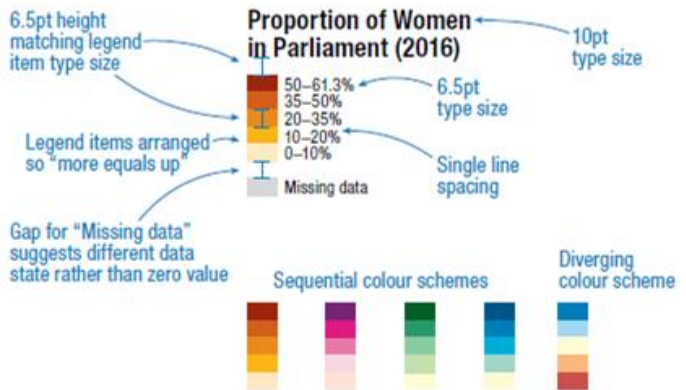
- country boundary (30K, 0.4 stroke width)
- figure arrowheads (100K, arrow 2, scale 80%, rounded cap)
- figure rounded rectangles (3pt corner radius)
- figure rounded rectangles outline (30K, 0.4 stroke width)

EXPORTS

- Place .AI files rather than image exports into layout
- Export a copy of maps/figures as .PDF on "high quality print" setting



Visual Style



Cartography often is described as both an art and a science. As an inter-governmental organization, the United Nations serves global *humanitarian* efforts, and its treatment of map design as an artistic process enables new ways to *humanize* the otherwise abstract statistical SDG indicator data.

As *visual art*, maps help us share our experiences of the world, promote empathy and compassion about uneven social and environmental conditions, and inform policy and politics for forging a sustainable future. As *visual culture*, maps reflect our interests and values, make us to confront our failures and prejudices, and reveal poten-

tial alternatives and opportunities.

The potential for artistic perspectives on cartography are numerous and ever expanding. One important way that artist elements enter every map design is through its *visual style*, or a cohesive set of design characteristics and qualities that reinforce rather than undermine the purpose of the map.

A visual style can be deconstructed into its elemental forms, colours, typefaces, and textures (Figure 2.14-1). *Form* describes the variable aspects of the geospatial linework, including its generalized detail (see Section 2.7) and line weights, cap and joint styles, and tapering. Colour and typography—de-

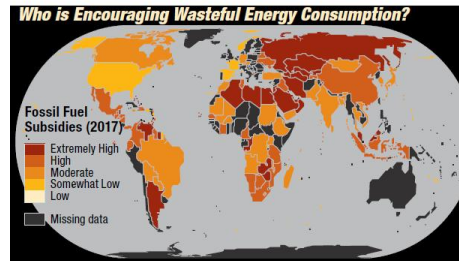
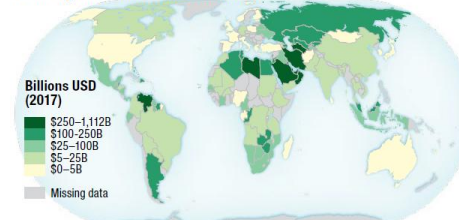


Figure 2.14-1: Visual styles in four web basemaps. **Left:** OpenStreetMap Mapnik uses detailed linework, complex colours and textures, and dense type, activating the audience and producing a sense of accuracy and trustworthiness in the basemap. **Left-centre:** Esri World Light Gray Canvas uses simplified linework and visual hierarchy along with muted colours, textures, and labels to deactivate the audience, making it a good basemap when additional data overlays should rise to figure (see Section 4.6). **Right-centre:** CARTO Dark Matter similarly uses simplified linework and sparse type and textures but with a darkened colour palette, setting an ominous, tense tone in the map. **Right:** Stamen Watercolor includes whimsical linework, bright colours with no type, and textures mimicking the spread of watercolours over paper, evoking a pleasant almost serene sense of beauty with the map.

Figure 2.14-2: Visual styles and emotion. Indicator 12.c.1 (2017) on per capita fossil fuel subsidies in USD is mapped in two different visual styles.

Top: The minimalist, authoritative style includes detailed linework and a soft colour palette. The map uses an arithmetic classification to emphasize low values in the left-skewed data distribution. Such a minimalist style likely evokes a pleasant, but deactivated emotional experience. **Bottom:** The alternative, creative style includes angular linework and bold colours against a black background, with the title framing the map purpose. Ordinal data are mapped with a quantile classification, placing more countries into the higher classes. Such a sensationalist style likely evokes an activated, unpleasant emotional experience, which may or may not be appropriate for the mapping context.

Per Capita Fossil Fuel Subsidies



fin in Section 2.10 and Section 2.11 respectively—greatly influence the map's visual style. Both colour and typography evoke strong but different connotations cross-culturally. Texture, a visual variable (see Section 2.9), describes the additional pattern fills and overlays that add visual complexity and often suggest a physical materiality or historical pastiche.

The visual style of the map has a major impact on the audience's emotional experience: both setting the mood as a pleasant or unpleasant topic while also engaging or boring the reader. Understanding the relationship between style and emotion helps cartographers identify

a visual style that matches the gravity of the mapped phenomena while avoiding blatant map propaganda. A minimalist, authoritative style often is applied when mapping the SDG indicators, such as many of the maps in this book (Figure 2.14-2, top), to assert neutrality and suggest data accuracy. However, no map design is perfect nor free from subjectivity (see Section 2.3), and alternative, creative visual styles may better promote awareness, generate dialogue, and stimulate action (Figure 2.14-2, bottom).

A visual style that works well in one mapping context may be illegible, misleading, or even harmful in others.